Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Files, routes, functions are going to be useful. Routes and pages are things I need to improve on.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Creating the functions because the math is simple and it's something I already know. Creating the UI and pages will be the hardest because I don't have much experience creating websites with python.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Creating the UI will likely require the most time and effort. The math should be implemented first.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Getting carried away with the UI and making everything perfect may interfere with my work. I will minimize my distractions by getting the main focus done first.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

1. Home

Choose Something

- Reaction Calculator
- How to use/help
- About

2. Reaction Calculator

Inputs:

Place Balanced Chem equat. here:
ex. $2H_2 + O_2 \rightarrow 2H_2O$

Place starting mass of reactant 1 (in grams): [ ]

Place starting mass of reactant 2 (in grams): [ ]
type "N" to skip

Place actual mass of product collected (in g): [ ]
type "N" to skip

3. ~~[illegible]~~ results

Balance reaction here

moles of each reactant:
reactant 1:
Mass: ##.## g
Moles: #.## mol

reactant 2:
Mass: a number or "N/A"
Mole: " " " "

limiting reactant:
reactant 1 or 2

excess reactant remaining:
##.## g

excess reactant:
reactant 2 or 1

Theoretical yield
#.## g of product

4. Help page

description of how to use the website

5. About.
describe all the formulas used and how I used them / how to use them on paper.

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**I will put that in the "about" section or the "how to use" section**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Chemicals: a list of a chemical dataclass, which will hold all the reactants/products in the reaction. It will change after every time the user changes reaction equations. Actual_yield: a float that will store the product mass. Changes after every user input theoretical_yield : a float that will store the calculated theoretical product mass. Computed after the stoichiometry calculations Percent_yield: a float that stores the calculated percent yield. Computed after the theoretical yield is known.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Chemical: Formula: the symbol it uses, set on creation Coefficient: how many there are, change based on the equation Mass: amount in grams Molar Mass: molar mass, set on creation Moles: based on the molar mass and mass
Routes (minimum 7)	home() calculate()

The names of your route functions	process_calulation() results() help() about() examples()
Pages (minimum 5) A brief description of the pages in your site	Home: what the user first see when opening the website, can select the other pages Calculate: where the user see the input form Results: where the user see the outputs based on the inputs from the input form in calculate Help: how to use the website About: what formulas I used in each calculation and how to do the calculations yourself
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Balanced equation: a textbox where the user inputs their balanced chemical equation Reactant mass: a textbox where the user inputs their reactants' masses Actual product yield: a textbox where the user inputs the product yield
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Check if input is a valid input: if invalid, it will return an error message Finding the limiting reactant: compares the mole-to-coefficient ratio of each reactant. If one reactant has the smallest ratio, the program identifies it as the limiting reactant and uses it to calculate the theoretical yield. Check valid percent yield: checks if the calculated percent yield is below, equal to, or above 100%. If the percent yield is greater than 100%, the program

	warns the user that there may be experimental error or contamination.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	A for loop that will loop in the chemicals list, changing each chemical.mole into the mass/molar mass

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.